

Unveiling Designers' Cognitive States in Early Engineering Design Stages: An Autoencoder-DNN Approach based on EEG Data*

Mingrui Li, Zuoxu Wang, Fan Li, and Jihong Liu

Abstract— Early engineering product design stages, such as innovative conceptual design and product form design, require intensive thinking by designers. During these tasks, designers experience constantly shifting cognitive states of either Trance, Concentration or Confusion. Accurately recognizing designer's cognitive states is a prerequisite task to provide assistance, e.g. design knowledge recommendation, to designers timely. Electroencephalogram (EEG) data is the external expression of designers' cognitive states. However, current research on applying EEG technology in engineering design scenarios lacks automatic selection of significant features from EEG information, and the connection with the design process is also limited. Faced with such issues, this study proposes a cognitive state recognition procedure for engineering design scenarios, including a recognition model and an experiment protocol. The Autoencoder-Deep Neural Network (DNN)-based recognition model can flexibly select significant features to achieve accurate cognitive state recognition, while the two-stage experiment protocol can collect standard cognitive state data and perform validation in the simulation of real design scenarios. The experiment results demonstrate the validity of the proposed recognition procedure, and confirmed that states of Concentration and Confusion can be utilized to determine whether designers need assistance.

I. INTRODUCTION

The design of engineering products requires extensive technical knowledge and specialized expertise [1]. In early design stages of complex engineering products, designers often struggle to conceive promising design solutions based on their own knowledge, which makes the design process arduous [2]. When encountering difficulties, designers typically tend to ponder deeply rather than immediately seeking assistance, leading to a slowdown in the design progress and affecting the coherence of design thinking. Therefore, automatically determining the state of designers is of great importance for enhancing the efficiency of engineering design.

Current methods of recognizing the designers' states include context-aware methods and physiological factor analysis. Context-aware methods relies on identifying the designer's situation by considering regular characteristics of the designer, the task, and the environment [3]. Nevertheless, this method needs pre-defined design scenarios, and it cannot

fully correspond to designers' variable internal states. Physiological factors, such as behavior, eye movements, and electroencephalogram (EEG) signals, can reflect designers' states more flexibly. Research based on behavior and eye movements can uncover some intentions of designers [4, 5], but they may not always fully align with design thinking due to susceptibility to external influences.

EEG data is the external expression of brain activity and can directly reveal the cognitive states of designers. However, in the field of engineering design, EEG data is predominantly utilized for evaluating or selecting completed design schemes statically, e.g., product sketch evaluation and selection. Current EEG data-based studies in engineering design field are still with limited connection to the design process, lacking a dynamic assessment on designers' cognition. Meanwhile, most studies use preprocessed EEG data directly for analysis, lacking the selection of significant features. This study aims to provide a cognitive state recognition procedure based on EEG data to support early engineering design stages. An Autoencoder [6]-Deep Neural Network (DNN, [7])-based approach is proposed to perform cognitive state recognition with automatic EEG feature selection, and the simulation of real design scenarios enables the investigation of designers' cognitive states during the design process.

The rest of this paper is organized as follows. The related works about EEG technology in engineering design is reviewed in Section 2. Section 3 introduces the proposed methodological procedure of cognitive state recognition in engineering design. Section 4 elaborates on the details of the experiment and discussed the results. The contributions and future research directions of this study are pointed out in Section 5.

II. RELATED WORKS

Existing studies that integrate EEG technology with engineering design mainly falls into three research paradigms, i.e., design scheme evaluation, design decision prediction and cognitive state recognition.

As the most commonly adopted paradigm, design scheme evaluation refers to linking EEG information to different levels of psychological states or integrating values from multiple indicators with different weights to generate an overall score,

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which is used to evaluate the design schemes of a product. The EEG data utilized in this paradigm generally comes from evaluators who are not directly involved in the design process, such as the users of the product. Jing et al. [8] extract design preferences from users' EEG data and correlate them with design features of the product to assist designers in obtaining design concepts that meet users' requirements. EEG data was associated with design preference word pairs to train a back propagation (BP) neural network for concept scheme evaluation in this study. Lai et al. [9] proposed a heterogeneous data fusion method that combines EEG data, eye-tracking data, and self-reports to evaluate product appearance design schemes. Lou et al. [10] employed the random forest algorithm as a classifier to map EEG information from customers to different implicit psychological states, and expert evaluations were integrated through cloud models to facilitate assessment of elevator design schemes. They further incorporated edge computing technology to enable collaborative evaluation by multiple decision-makers [11].

Design decision prediction refers to identifying decision-related information from EEG data for the automatic selection of design schemes. Different from scheme evaluation, this paradigm involves the use of designers' EEG data, thus having a closer relation with the design process. Wang et al. [12] predicted the design decision-making by integrating features from EEG and eye-tracking data through a multimodal fusion strategy, reducing the impact of subjective factors unrelated to actual preferences on decision-making. Cao et al. [13] employed Common Spatial Patterns (CSP) to extract spatial features from EEG in different frequency bands, then trained a SE-Net-based convolutional neural network (CNN) as a selection intention discrimination model to predict object selection in the computer-aided design (CAD) process. Their research provided a new idea to address the Midas touch problem (visual attention may not reflect selection intension) in traditional item selection approaches merely based on gaze data.

However, both of the first two research paradigms require the formation of relatively concrete design schemes before applying EEG technology, which are then available for users

or designers to evaluate or select from. These two paradigms still cannot directly support designers during the design process. Therefore, how to fully integrate EEG data into the formation process of design schemes is a crucial issue in enhancing its utility in engineering design.

The objective of cognitive state recognition is to utilize the EEG data of designers in design process to identify cognitive states in different time periods, providing assistance in diverse aspects of engineering design, such as cognitive pattern research and knowledge recommendation. Liang et al. [14] proposed a knowledge requirement awareness framework that considers design context and EEG information. They used cognitive states reflected by EEG, combined with the design context, to activate specific knowledge requirements in different design tasks, thus enabling the recognition of knowledge requirements. Nevertheless, research focusing on this paradigm is still relatively limited in quantity, lacking sufficient depth and comprehensiveness. This study recognizes cognitive states in an engineering design scenario with the assistance of Autoencoder and DNN models, with the aim of providing a procedure to investigate the cognitive patterns of designers when operating different design elements.

III. METHODOLOGY

This section elaborates on the methodological process of cognitive state recognition in early-stage engineering design scenarios. The overall cognitive state recognition procedure is illustrated in Fig. 1. We first categorize cognitive states into Trance, Concentration and Confusion, and utilize EEG data tagged with these three states to train a DNN model as a cognitive state recognition model. During this process, we incorporate an Autoencoder to automatically extract significant features from EEG data, thereby enhancing the accuracy of recognition. Then, we use eye gaze data to identify the design elements that designers focus on in different time periods, and employ the cognitive state recognition model to identify cognitive states from EEG data in these time periods, thus obtaining the cognitive states of designers when processing different design elements.

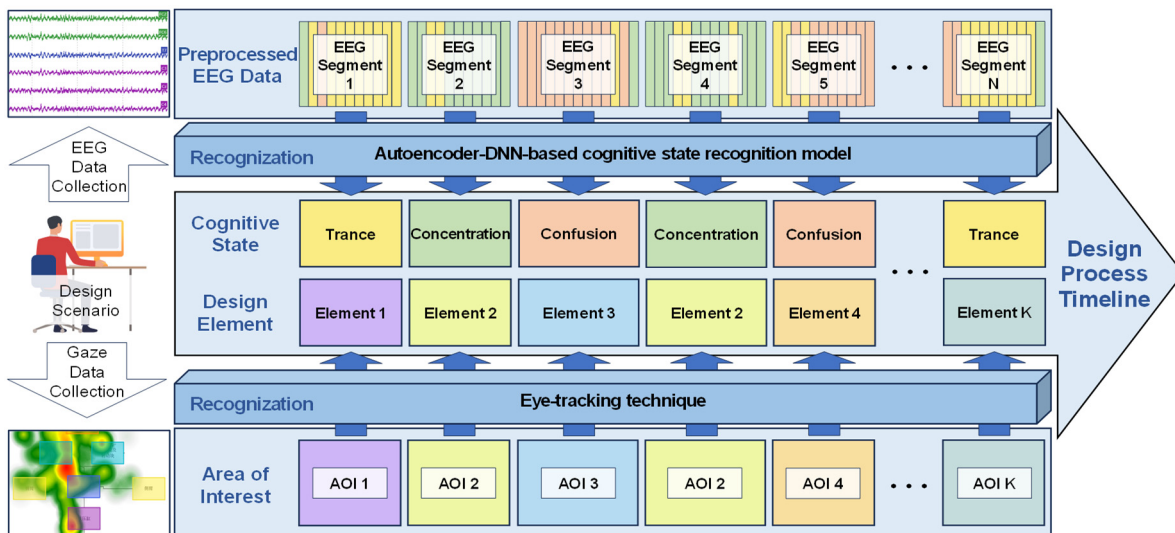


Figure 1. Procedure of cognitive state recognition in engineering design

A. Cognitive state recognition model establishment with EEG feature selection

In the engineering design process, designers experience different cognitive states while operating various design elements. In this study, we predefined three types of cognitive states, namely *Trance*, *Concentration* and *Confusion* based on pre-experiments to categorize and represent the ever-changing cognitive states of designers. Trance is a common cognitive state in lengthy design processes, wherein designers enter a condition of absent-mindedness. In the Concentration state, designers have sufficient ideas about the current design element and perform related operations attentively. The Confusion state often arises when designers encounter difficulties with a particular design element, causing them to contemplate solutions.

To obtain a model capable of recognizing the three cognitive states from EEG signals, we proposed a three-stage approach based on Autoencoder and DNN, as shown in Fig. 2.

In the first stage, preprocessed EEG data tagged with the three cognitive states is converted into vectors through feature extraction and reshaping. Feature extraction involves power spectral density (PSD) feature extraction, band power calculation and band selection. As a commonly used EEG feature, PSD values across different channels reflect the activation levels of different brain regions [15]. In this study, we calculated the PSD features for individual channels using the Welch's method and averaged them across the delta (1-4 Hz), theta (4-8 Hz), alpha (8-14 Hz), beta (14-30 Hz), gamma (30-49 Hz). Subsequently, we performed integration of these PSD features over the frequency range of each band to obtain the band power, based on:

$$Power = \int_{band_range} PSD(f) df \quad (1)$$

where f refers to the frequency of the EEG signal and $PSD(f)$ is the PSD value at this frequency. The band power values for all channels and all the frequency bands could be organized into a matrix, which serves as the feature representation of the EEG data segment. However, in some frequency bands, EEG features unrelated to the engineering design task can significantly hinder the recognition of cognitive states. To mitigate this issue, we trained the cognitive state recognition model using different numbers and

combinations of frequency bands and selected the final combination by comparing the models' accuracy on the test set. After this step, irrelevant data in the feature matrix could be removed, and the remaining data could be reshaped into a feature vector.

In the second stage, the feature vectors are further refined with the assistance of an Autoencoder. During an engineering design process, not all brain regions are actively involved in. Therefore, even though irrelevant frequency bands have been removed in the first stage, data from some channels within the remaining bands still does not represent cognitive state features, which adversely affects cognitive state recognition. To rapidly identify the most significant features from the feature vector and enhance the accuracy of the recognition model, an Autoencoder is employed for automatic feature selection. The Autoencoder can compress input data into a lower-dimensional representation and decode it back to the original dimension for reconstruction, and then compare it with the input data to calculate loss. After an unsupervised learning process, the indices of significant features could be extracted from the weights of the encoding layer. Consequently, the feature vector could be dimensionally reduced, retaining only the dimensions corresponding to these indices, as the final representation of the EEG data segment.

The third stage is the training process of the cognitive state recognition model. We use the EEG feature vectors derived from the first two stages, along with corresponding cognitive state tags, to train a DNN model on a classification task, and the trained model could be utilized to recognize cognitive states from EEG data.

B. Cognitive state recognition in real design scenario

In real design scenarios, performing cognitive state recognition requires segmenting the EEG data based on the time in processing different design elements. To determine the design elements that designers focus on at different times, this study adopted eye-tracking technology for Area of Interest (AOI) investigation. Specifically, we designate different kinds of AOIs on the design page to represent different design elements. As designers operate design elements, the shapes and positions of the corresponding AOIs also change accordingly. The gaze data obtained through eye-tracking technology can be presented as the start and end times on each fixation on every AOI, thus providing several time periods linked to distinct design elements.

The EEG data of designers in this process is also a temporal sequence, thus it could be divided according to the above-mentioned time periods to obtain EEG segments corresponding to different design elements. After being processed by the cognitive state recognition model, these EEG segments could be tagged with different cognitive states. For a particular design element, the predominant tag among all segments will be considered as the designer's cognitive state regarding the design element.

C. Experiment protocol

Fig. 3 illustrates the procedure of the experiment protocol, which consists of two stages. The first stage aims to collect EEG data under the three cognitive states, i.e., Trance, Concentration and Confusion, to build the dataset for model training. There are 11 rounds in total in this stage of the

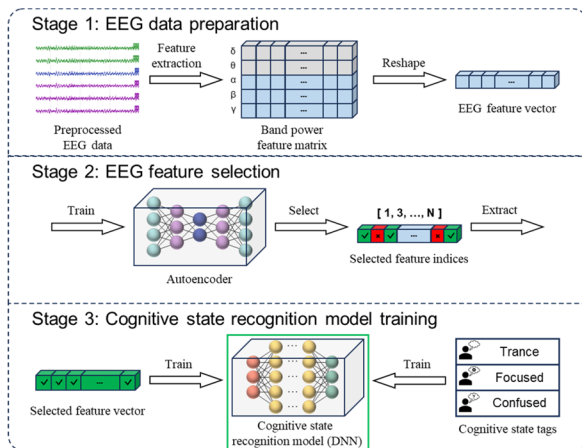


Figure 2. Cognitive state recognition model establishment process

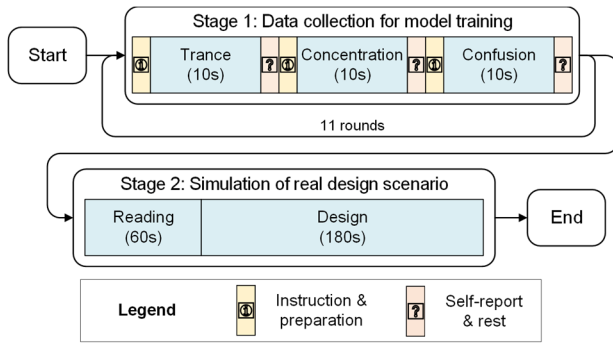


Figure 3. Experiment protocol for cognitive state recognition

experiment, with the first round used to familiarize participants with the experiment process and therefore excluded while constructing the dataset. In each round, participants are required to complete tasks corresponding to the three cognitive states, with each task lasting 10 seconds. For the Trance state, a page identical to the design interface except that all text is replaced with 'x' is displayed on the screen. Participants are instructed to gaze at the blank area and let their minds wander. For the Concentration state, participants are asked to finish an easy task of arranging design elements guided by a description. For the Confusion state, participants are presented with a partially completed product structure diagram on the screen, with some elements still outside the working area. They need to imagine where the unfinished elements should be placed solely based on the given design problem, and its difficulty ensures participants do not generate concrete ideas in a short period of time. Before each task, participants need to read an instruction and start by clicking the mouse when they are ready. After completing the task, they are asked to verbally report whether they followed the instruction and may take a rest if needed.

The second stage aims to simulate an engineering design scenario and collect participants' EEG data when operating different design elements, to validate the effectiveness of the cognitive state recognition model in practical application. Eye gaze data was also recorded in this stage to identify the design elements participants were focusing on at different moments. In this stage, participants are required to create schematic diagrams of the target product based on a design problem and some given design elements. Considering participants' limited familiarity with the structure of aircraft landing gear, a 60-second reading session is arranged prior to the design process to equip participants with some introductory knowledge, with no requirement of full memorization. Then, within a timeframe of 180 seconds, participants are asked to drag the given design elements into the working area and connect them to form the schematic diagram. After all participants completed the experiment, domain experts are invited to assess whether each design element in the schematic diagrams they created is correctly arranged.

IV. EXPERIMENTS AND RESULTS

A. Data collection & preprocessing

The experiment setting is shown in Fig. 4. A total of 12 participants were recruited through a questionnaire. They were all master's or doctoral students majoring in mechanical engineering, and it was ensured that they were in good physical

and mental condition during the experiment. Before the experiment began, each participant signed a consent form. The raw EEG data was collected using ErgoLab EEG Device (developed by Kingfar Co., Ltd.) that includes a 32-channel semi-dry EEG cap and an amplifier, and the electrodes were located according to the international 10-20 system, as illustrated in Fig. 5. A reference electrode was positioned on the left mastoid. The sampling rate of the device was set to 256 Hz, and the bandwidth was 1-49 Hz. The eye gaze data was collected via Tobii Pro Fusion Eye Tracker (developed by Tobii Technology), with a sampling rate of 30 Hz. Both the EEG data and the eye gaze data were synchronously recorded by ErgoLab software (also developed by Kingfar Co., Ltd.). The design process was configured to create a schematic diagram of the target product structure using Microsoft Visio software. All the tools and devices were reset and calibrated before the initiation of the experiment for each participant.

The experiment was conducted following the proposed two-stage experiment protocol. The design problem was extracted from Patent No. CN220315286U, which relates to a kind of landing gear linkage assembly. The introductory reading material was a segment of the patent abstract, and a total of 10 design elements with 8 types were provided. The design interface is shown in Fig. 6, with the preceding reading material and a predefined feasible solution (both hidden in this session) also demonstrated. The blocks of different colors in Fig. 6 represent the AOIs corresponding to different design elements, which were also hidden in this session.

After the completion of the experiment, a set of data preprocessing steps was applied to the raw EEG data with the support of EEGLAB toolbox in Matlab. Firstly, the raw EEG data was transformed into relative values based on reference electrode data. Secondly, invalid EEG data segments were removed, which originated from the poor contact problems of

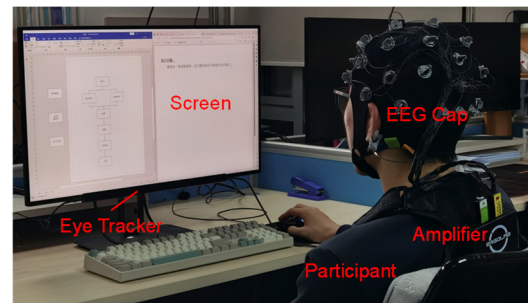


Figure 4. Experiment setting in this study

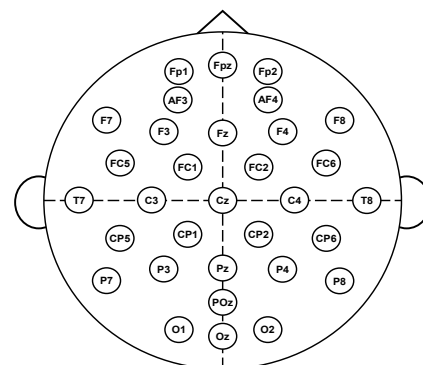


Figure 5. EEG channel locations

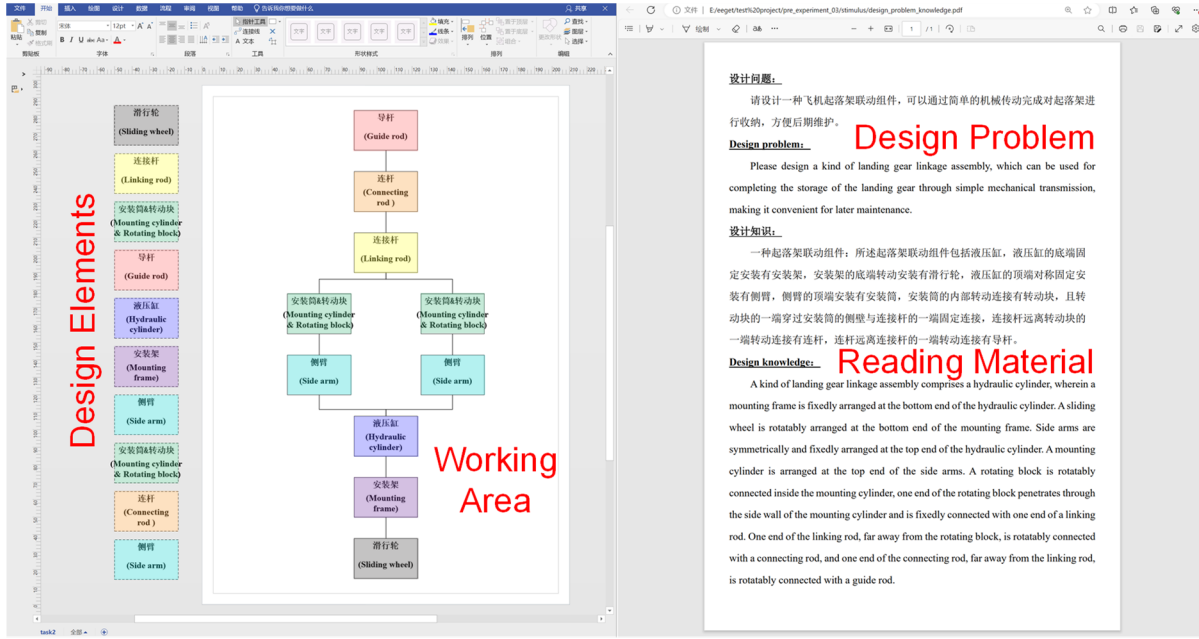


Figure 6. Illustration of the design interface, the reading material, a predefined feasible solution and the AOIs

the equipment and tasks that were reported as not following the instructions by participants themselves. Thirdly, we interpolated the few invalid channels based on the data from neighboring channels. Lastly, the EEG data was filtered with a bandpass filter ranging from 1 to 49 Hz.

To construct the dataset for training the cognitive state recognition model, we further trimmed and augmented the preprocessed EEG data. For the 10-second EEG segments obtained from the first stage of the experiment, we divided the data from the last 8 seconds into 1-second segments using a sliding window with a quarter-second step size. Finally, a dataset containing 8584 valid EEG data segments tagged with cognitive states was constructed.

B. Cognitive state recognition model training & evaluation

As a preparatory step for model training, we performed feature extraction on the entire dataset. After multiple rounds of model training and comparison, the data from the alpha, beta and gamma frequency bands were selected and represented as a feature matrix (32 channels $\times$ 3 bands), and then further reshaped into feature vectors of length 96.

Both the Autoencoder and the DNN model were trained on the PyTorch framework. The dataset was divided into training, validation and test sets in an 18:1:1 ratio. For the Autoencoder, we set the dimensions of the hidden layer and the encoding layer to 80 and 64, respectively, and trained for 20 rounds with a learning rate of 0.0004. With the Autoencoder, 90 significant features were selected and formed into final EEG feature vectors. For the DNN model, we configured 3 hidden layers with 1000 units, trained for 2000 epochs with a learning rate of 0.0001, and obtained the best model from the 1583rd epoch.

To comparatively validate the advantage of our Autoencoder-DNN-based approach, we used four machine learning models, i.e., K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Decision Tree (DT) and Random Forest (RF), as well as a Convolutional Neural Network (CNN) model, as baselines and conducted experiments with and

without the Autoencoder. The accuracy in classifying the three cognitive states is employed as the evaluation metric, which is calculated by:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (2)$$

where TP , FP , TN and FN refer to true-positive, false-positive, true-negative and false-negative, respectively.

As Table I shows, with the Autoencoder selecting significant features, the accuracy of all models has increased, and our combination of Autoencoder and DNN achieved the highest accuracy of 88.4%. This result also demonstrates that our recognition model can effectively classify different cognitive states.

C. Results & discussion in the design scenario

Since the cognitive state recognition model was trained on 1-second EEG segments, we divided the EEG data recorded during each participant's gaze at each AOI in the second stage of the experiment into 1-second segments as well, and gaze segments that were less than 1 second were excluded from consideration. A total of 355 EEG segments were obtained and organized according to their corresponding types of design elements. After completing the recognition of cognitive states, we recorded the predominant cognitive states of each participant when focusing on each type of design element and analyzed them in conjunction with the assessments of the participant's arrangement for each design element by domain experts.

The analysis results presented in Table II reveal the relationship between different cognitive states in design scenario and the validity of design schemes. When designers are in the Trance state, an equal number of correct and incorrect design elements are observed, confirming that this state may originate from either the design element not

TABLE I. ACCURACY OF THE MODELS IN EEG CLASSIFICATION

Model	Without Autoencoder	With Autoencoder
SVM	49.0%	50.6%
DT	58.8%	59.6%
RF	78.6%	80.8%
KNN	82.7%	83.2%
CNN	85.6%	85.8%
DNN	84.5%	88.4%

TABLE II. DESIGN VALIDITY IN DIFFERENT COGNITIVE STATES

Cognitive state	Result	
	Correct	Incorrect
Trance	12	12
Concentration	14	7
Confusion	5	26

requiring much consideration or the designer having no idea about at all. When designers are in the Concentration state, the majority of design elements are arranged correctly, confirming that designers have clear ideas and requiring little assistance. When designers are in the Confusion state, the arrangements of only a very slight number of design elements were deemed acceptable, demonstrates that this cognitive state truly arises from designers encountering difficulties and struggling to come up with appropriate solutions. In such circumstances, it is necessary to provide knowledge support to designers regarding the current design element.

V. CONCLUSION

This study presents a procedure for recognizing designers' cognitive states in early engineering design stages as well as an Autoencoder-DNN-based recognition approach. The contributions of this study are summarized as the following three points.

- An Autoencoder-DNN-based cognitive state recognition approach is proposed. The Autoencoder can automatically identify significant features from EEG data through unsupervised learning, and the multi-layer structure of the DNN model can accurately recognize different cognitive states based on these features.
- A two-stage EEG experiment protocol towards cognitive state collection and recognition is introduced. Specifically, standard cognitive state data with tags was collected in the first stage, and then used to train a high-quality cognitive state recognition model. The validation of the trained model was conducted under the simulation of real design scenarios. This experiment protocol enables a more realistic reflection of the ever-changing cognitive states of designers during the design process.
- Three cognitive states of designers in engineering design scenarios were investigated. The findings from the experiment suggests that the states of Concentration and Confusion can be used to determine whether designers need assistance.

However, the experiment in this study only investigated and validated the proposed procedure on a theoretical design scenario, while engineering design involves various other types of scenarios, and that causes certain limitations. Meanwhile, the systematicness of the experiment also has room for improvement. Our future work will focus on a broader range of design scenarios and the optimization of experimental design, with the goal of providing a widely applicable method to improve the efficiency of engineering design.

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